When Finite Differences Flip the Sign of Second-Order Generative Geometry: A Non-Recoverability Result and an Exact Instrument

Anonymous authors
Paper under double-blind review

Abstract

Finite-difference estimates of second-order geometry in generative models are unreliable, and we show this in two distinct ways. On Stable Diffusion (Rombach et al., 2022), a finite-difference estimate of a curvature ratio destabilizes the sign of its rank correlation with the first-order singular value: exact forward-mode automatic differentiation gives a consistent -0.73 (negative on five of six seeds), while the finite-difference estimate of the same quantity scatters in sign across seeds and averages -0.18. Separately, the naive second difference is *non-recoverable*: truncation and floating-point cancellation cannot be made small at the same step, leaving a relative-error floor of order $\sqrt{\varepsilon_{\text{mach}}}$ that single precision makes large. On a StyleGAN generator (Karras et al., 2019) this floor is at least 45% at every fixed step and in both single and double precision, whereas the first-order finite difference of the *same* code agrees with forward-mode AD to 2.4%, the asymmetry that makes the failure specific to second order. We use composed forward-mode AD as a step-size-free estimator of the higher-order pushforward (the exact derivative of the implemented computational graph, up to floating-point arithmetic), and audit a representative set of latent-geometry methods to locate where the $/\varepsilon^2$ estimate is and is not used. We do not claim any published result is wrong; most methods avoid the regime by construction. We characterize when finite differences fail for second-order generative geometry and supply a step-size-free replacement.

1 Introduction

Let $G : \mathbb{R}^d \to \mathbb{R}^{3HW}$ be a trained generator. Treating the image set $\{G(z)\}$ as a manifold pulled back through G, a line of work studies its geometry: the pullback metric $M_z = J_z^\top J_z$ and its singular vectors (the local latent basis), geodesics between latent codes, and the curvature of the generated manifold (Arvanitidis et al., 2018; Shao et al., 2018; Chen et al., 2018; Park et al., 2023; Saito & Matsubara, 2025a). The first-order objects J_z and M_z are routine to compute. The second-order objects are not: the curvature, the second-order pushforward $\partial^2 G/\partial\alpha^2$ along a path, and the metric derivative $\partial_z M_z$ that appears in the geodesic ODE. These are where the numerical method can silently flip a result.

We start with a measurement (Section 2). On Stable Diffusion, a finite-difference estimate of a second-order curvature ratio destabilizes the sign of a rank correlation that the exact forward-mode estimate reports consistently across seeds. The finite-difference route here is a central difference of the exact first-order JVP; its instability is empirical, and we attribute it to the nonlinearity of the deep denoising chain rather than to the cancellation mechanism below. A practitioner who reached for it would draw the wrong qualitative conclusion.

A separate and stronger result concerns the naive second difference (Section 3). A central finite difference is well conditioned for a first derivative but non-recoverable for a second: truncation and floating-point cancellation trade off and cannot be small together, leaving a relative-error floor of order $\sqrt{\varepsilon_{\text{mach}}}$. In the single precision that GANs and diffusion U-Nets run in that floor is large, and on a strongly nonlinear, non-smooth generator it is at least 45% at every step and in both precisions (Section 4). The first-order finite difference of the same code stays accurate; this asymmetry is our validation control.

We contribute two results: a reproducible sign-instability of finite-difference second-order geometry on a real diffusion pipeline, and a non-recoverability account of the naive second difference, with the first/second-order asymmetry as its practitioner-facing consequence. We also supply a step-size-free instrument for the higher-order pushforward, built by composing forward-mode AD, and an audit that locates which parts of the literature use the $/\varepsilon^2$ estimate and which avoid it. The contribution is a precise account of a real numerical hazard and the fix, not a claim that any published method is wrong; the audit finds that most route around the regime by construction (Section 7).

2 The sign-inversion phenomenon

Setup. The exact estimator is forward-mode automatic differentiation composed with itself: nesting two Jacobian–vector products (JVPs) returns the second-order directional derivative $\partial_\alpha^2 G$ with no step size (Section 5). Throughout, "exact" means the derivative of the implemented computational graph, up to floating-point arithmetic, not a finite-difference approximation. The finite-difference baseline here is a central difference of the exact JVP at step ε, that is, a *first* difference of $\alpha \mapsto \partial_\alpha G$ rather than the naive second difference of Sections 3–4; it is the milder, better-conditioned route, which makes its instability below the more notable. We run both on Stable Diffusion v1.5 (Rombach et al., 2022) with a 20-step DDIM schedule (Song et al., 2021) and classifier-free guidance (Ho & Salimans, 2022), in the U-Net h-space. From a base latent we draw K=16 random h-space probes, build the Jacobian J restricted to their span, take its top k=12 right-singular directions u_i with singular values σ_i, and along each measure a second-order curvature ratio

$$\rho_i \;=\; \left.\frac{|\,\partial_\alpha^2 G(z+\alpha u_i)\,|}{|\,\partial_\alpha G(z+\alpha u_i)\,|}\right|_{\alpha=0},$$

where $|\cdot|$ is the mean absolute value over output pixels.

We then ask a standard question: does curvature co-vary with the first-order singular value σ_i? We compute Spearman(σ, ρ) with the second derivative in ρ taken both ways, exact and finite-difference.

Over 6 seeds, with the top 12 directions each, the finite-difference estimate is sign-unstable where the exact estimate is consistent (Table 1). The exact estimate gives a strong negative correlation on five of six seeds, clustered in $[-0.94, -0.82]$, with seed 4 a near-zero outlier at -0.03; the mean is -0.73. The finite-difference estimate of the same ratio scatters in sign across seeds (positive on two) and averages -0.18, attenuated toward zero. The exact estimate is more negative than the finite-difference estimate on all six seeds, a consistent ordering; the corresponding Wilcoxon signed-rank $p = 0.031$ is the smallest value n=6 can produce and depends on the near-zero seed 4, so we read it as consistent ordering rather than a powered test. The finite-difference estimate is sign-unstable at every step in the grid $\{0.1, 0.05, 0.01, 0.001\}$, with means between -0.08 and -0.34, so no step reproduces the clean -0.73. A study built on the finite-difference route would report "no clear relationship," the opposite of what the exact second-order estimate finds.

seed	0	1	2	3	4	5	mean
exact composed JVP	−0.85	−0.83	−0.92	−0.94	−0.03	−0.84	−0.73
FD-of-JVP(ε=0.05)	+0.18	−0.55	−0.78	−0.40	+0.55	−0.07	−0.18

Table 1: Spearman(σ, ρ) on Stable Diffusion, per seed. The exact second-order estimate is consistently negative; the finite-difference estimate of the identical quantity changes sign across seeds and averages near zero. We define ρ to expose the method dependence; the point is that the two numerical routes disagree on its sign, not the specific value of any prior work.

3 Why finite differences are non-recoverable for a second derivative

Proposition 1 (Second-order non-recoverability)**.** *Let $f \in C^4$ near α with $f^{(4)}$ bounded and $f''(\alpha) \neq 0$, evaluated in floating point with unit roundoff $\varepsilon_{\text{mach}}$ under a standard rounding model (Higham, 2002). The* *central second difference* $D^2_\varepsilon f = \big(f(\alpha+\varepsilon) - 2f(\alpha) + f(\alpha-\varepsilon)\big)/\varepsilon^2$ *has relative error*

$$\frac{|D^2_\varepsilon f - f''(\alpha)|}{|f''(\alpha)|} \lesssim C_T\,\varepsilon^2 + C_R\,\varepsilon_{\mathrm{mach}}/\varepsilon^2, \qquad C_T = \frac{|f^{(4)}(\alpha)|}{12\,|f''(\alpha)|},\ \ C_R = O\Big(\frac{|f(\alpha)|}{|f''(\alpha)|}\Big),$$

whose minimum over the step is of order $\sqrt{C_T C_R\,\varepsilon_{\mathrm{mach}}}$*, attained at* $\varepsilon^\star \sim \varepsilon_{\mathrm{mach}}^{1/4}$*; no step does better. The central first difference, by contrast, attains a floor of order* $\varepsilon_{\mathrm{mach}}^{2/3}$*. For a vector map* $G : \mathbb{R}^d \to \mathbb{R}^{3HW}$ *the bound holds componentwise, so a single global step is governed by the worst component.*

The argument is elementary; we give it because the consequence is easy to miss in practice. The central second difference satisfies

$$D^2_\varepsilon f - f''(\alpha) = \underbrace{\tfrac{\varepsilon^2}{12} f^{(4)}(\alpha) + O(\varepsilon^4)}_{\text{truncation}} + \underbrace{O\big(\varepsilon_{\mathrm{mach}}\,|f|/\varepsilon^2\big)}_{\text{cancellation}},$$

and the two terms cannot be small together: minimizing their sum gives $\varepsilon^\star \sim \varepsilon_{\mathrm{mach}}^{1/4}$ and the floor above (full derivation in Appendix C). The exponent $\frac{1}{2}$ is intrinsic to the operation; the constant carries the generator's nonlinearity $|f^{(4)} f|/f''^2$, which is large for a deep network. In double precision the floor is about 10^{-8} in the benign (constant near one) case. In the single precision generators use ($\varepsilon_{\mathrm{mach}} \approx 1.2 \times 10^{-7}$ for fp32) it is about 3×10^{-4} benign, and far worse for a strongly nonlinear G, where Section 4 measures $\geq 45\%$. The floor is a property of the operation in fixed precision, not of any architecture, so the hazard is general; we demonstrate its magnitude on two generator families.

The first derivative behaves differently. The central first difference $\big(f(\alpha+\varepsilon) - f(\alpha-\varepsilon)\big)/2\varepsilon$ has truncation $O(\varepsilon^2)$ and cancellation $O(\varepsilon_{\mathrm{mach}}|f|/\varepsilon)$, a much milder floor $\varepsilon_{\mathrm{mach}}^{2/3}$ with a wide usable window. First-order Jacobians are therefore safe to finite-difference while second-order curvature is not. This is also the control we use throughout: wherever the first-order finite difference agrees with the exact JVP, the pipeline is correct, and any second-order disagreement is real rather than a bug.

To check the constants against ground truth, let $f(\alpha) = [\,e^{0.7\alpha}, \sin 3\alpha, (\alpha+0.4)^3, \tanh 2\alpha, \alpha^2\cos\alpha\,]$, whose derivatives are known in closed form. At $\alpha = 0.3$ in double precision (Table 2) the central second-order relative error is U-shaped: it bottoms at 8.6×10^{-9} near $\varepsilon = 10^{-4}$ and diverges to 0.29 by $\varepsilon = 10^{-8}$. The composed JVP reaches 2.5×10^{-17}, near machine precision, so the best finite difference is 3.5×10^8 times worse. The measured best-FD error matches the Proposition 1 bound with the 1/12 truncation constant, to within the rounding model.

ε	central FD 2nd-order rel. err.	
10^{-1}	7.9×10^{-3}	(truncation)
10^{-2}	8.0×10^{-5}	
10^{-4}	8.6×10^{-9}	(best)
10^{-6}	3.0×10^{-5}	
10^{-8}	2.9×10^{-1}	(cancellation)
composed JVP	2.5×10^{-17}	(exact)

Table 2: Second-order accuracy on the analytic toy at $\alpha = 0.3$, double precision. Finite differences never approach the exact composed JVP.

4 Quantification on a real generator

On a real generator the failure is worse than the toy predicts, and for a second reason. We compare the naive second-difference curvature a practitioner would write, $D^2_\delta G = \big(G(z+\delta u) - 2G(z) + G(z-\delta u)\big)/\delta^2$, against the exact composed JVP $\partial^2_\alpha G(z+\alpha u)$ on a StyleGAN-bedroom generator at 256×256, over 6 directions (3 annotated attributes, 3 random) and 16 samples, sweeping δ (Figure 1, Table 3). The exact curvature is a

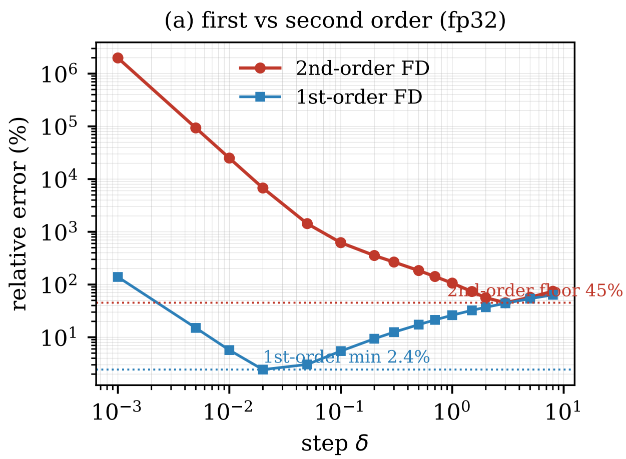

Figure 1: The non-recoverability of Proposition 1 on a StyleGAN generator. (a) Relative error of the finite-difference curvature versus step δ in fp32: the first-order difference (blue) has a usable minimum near 2.4%, while the second-order difference (red) never falls below 45%. The exact composed JVP has error 0 at any δ. (b) The second-order error in fp32 versus fp64: double precision removes the small-step cancellation but leaves the best-step floor unchanged at 47%, so no precision recovers the curvature.

definite, step-free number whose magnitude varies by direction (per-direction means span 0.03 to 6.6), so we report relative errors per direction. Holding a single step across all samples, as a practitioner does, the mean per-direction finite-difference relative error never drops below 45.2% (best step $\delta = 3$, bracketed on both sides; the median is 38% and the distribution is heavy-tailed, so 45.2% is a mean of per-sample ratios, not a ratio of the displayed means). It is 267% at $\delta \approx 0.3$ and explodes in fp32 as $\delta \to 0$. Even an oracle per-sample step gives a median best error near 9%, against the composed JVP's zero.

Precision does not rescue it. On a matched subset (3 directions $\times$ 8 samples) rerun in both precisions, the best-step floor is 47.1% in fp32 and 47.1% in fp64, unchanged, and the mechanism is not cancellation alone. At usable steps the error is truncation and is precision-blind. At tiny steps fp32 adds catastrophic cancellation, but fp64, which removes that, instead exposes the generator's non-smoothness: across a LeakyReLU kink the chord-scale second difference does not converge to the pointwise pushforward, and the fp64 error peaks near 3000–3500% at small steps. So no step and no precision recovers the exact second-order pushforward, which the composed JVP returns directly.

The control confirms the instrument is correct. The first-order central difference of the same pipeline agrees with the exact first-order JVP to 2.4%, so the second-order failure is genuine.

5 The exact instrument

Forward-mode AD evaluates, for a map f and tangent v, the Jacobian–vector product $\partial f(x)[v]$ alongside $f(x)$ in one pass, without forming the Jacobian. With $f = G$, $x = \gamma(\alpha)$, and $v = u$ this gives the exact first-order pushforward. The second-order term is the directional derivative of the first, so composing forward mode with itself,

$$\partial_\alpha^2 G(\gamma(\alpha)) = \partial_\alpha\big[\partial G(\gamma(\alpha))[u]\big][u],$$

is the exact derivative of the implemented graph (up to floating-point arithmetic) and is evaluated by nesting two JVP calls: in PyTorch, `torch.func.jvp` of a function that itself returns a JVP (Paszke et al., 2019). No full Jacobian or Hessian is materialized, so memory is dominated by a forward evaluation and tangent propagation rather than by the output dimension $3HW$, and the same nesting yields any order in a constant number of passes. Composing forward-mode passes for higher-order directional derivatives is standard algorithmic differentiation (Griewank & Walther, 2008); our use of it is as the step-size-free ground truth against which the finite-difference practice fails. The instrument is generator-agnostic: we ran

step δ	$\lvert D_\delta^2 G\rvert$	2nd-order rel. err.	1st-order rel. err.
8	7.1×10^{-3}	74.5%	63.8%
3	2.7×10^{-2}	45.2% (floor)	44.1%
1	9.9×10^{-2}	106.4%	26.2%
0.3	3.7×10^{-1}	266.9%	12.5%
0.1	1.06	622.8%	5.5%
0.02	4.05	6.8×10^{3}%	2.4% (1st-order min)
10^{-3}	4.09×10^{2}	2.0×10^{6}%	139%
composed JVP	1.17	exact (step-free)	—

Table 3: StyleGAN-bedroom, fp32, 6 directions ×16 samples. The relative-error column is the mean per-direction relative error (median 38% at $\delta = 3$; heavy-tailed); $\lvert D_\delta^2 G\rvert$ is the aggregate magnitude, shown for scale and not the quantity the relative error is computed from. The naive second difference never gets within 45.2% of the exact curvature at any fixed step, while the first-order difference of the same code agrees to 2.4%. A separate precision-control subset (3 directions ×8 samples; Appendix E) gives a best-step floor of 47.1% in *both* fp32 and fp64, so the floor is truncation at usable steps and generator non-smoothness at tiny steps, not a single-precision artifact.

it unmodified on StyleGAN1 and 2 (Karras et al., 2019; 2020) and on a latent-diffusion U-Net (Rombach et al., 2022).

6 Related work and exposure audit

The finite-difference step-size tradeoff is classical numerical analysis (Press et al., 2007), and higher-order forward-mode AD is standard (Griewank & Walther, 2008); what has been missing is an account of where the tradeoff bites in the differential geometry of generative models, and how often. We classify a targeted, representative set of latent-geometry methods, those most directly adjacent to curvature, geodesics, and the pullback metric, by how each obtains (or avoids) a second-order or geodesic quantity (the per-paper test is in Appendix G); this is a representative audit, not an exhaustive survey. Table 4 gives the result. The explicit $/\varepsilon^2$ second-difference of Section 3 is the obvious estimate for a practitioner adding a curvature measurement, but most published methods never form it. They keep the metric first order, replace the geodesic ODE with a discretized energy whose exact gradient is a well-conditioned discrete Laplacian rather than a $/\varepsilon^2$ curvature, use analytic Christoffel symbols, or differentiate the energy with autodiff (Yang et al., 2018); others compute no second derivative at all (Park et al., 2023; Chen et al., 2018; Kwon et al., 2023). The closest adjacent practice that does use a finite difference is the first-order difference of a network derivative, such as a score-Jacobian–vector product; it lies on the well-conditioned $\varepsilon_{\text{mach}}^{2/3}$ side of Section 3 and so is not exposed in our sense.

The field is mostly safe because the canonical formulations never form the estimate, but that safety is a property of those formulations rather than a law. As curvature- and geodesic-level analysis of diffusion models grows, for instance the energy-and-score geometry of Saito & Matsubara (2025a;b) or any second-order extension of the first-order bases of Park et al. (2023), the $/\varepsilon^2$ estimate becomes the obvious next step. Table 4, the asymmetry, and the exact instrument are meant as a reference for that emerging work.

Guidance. Finite-difference first-order quantities freely: choose a step near $\varepsilon_{\text{mach}}^{1/3}$ and expect a few digits. Never finite-difference a second derivative. In single precision no step gives even one reliable digit on a deep generator (Section 4), and the error can flip the sign of a downstream conclusion (Section 2). Use composed forward-mode AD: it is exact, costs one extra forward pass per order, and adds no memory. It also gives a free check. If the first-order finite difference and the JVP disagree, the pipeline is wrong; if they agree, any second-order disagreement is real.

Paper	2nd-order quantity	how obtained
Shao et al. (2018)	discretized geodesic energy	exact discrete-energy gradient (Laplacian), coarse T; no $/\varepsilon^2$
Saito & Matsubara (2025a)	geodesic (energy)	first-order score-differences $\|s(x_{i+1})-s(x_i)\|^2$; no $/\varepsilon^2$
Arvanitidis et al. (2018)	geodesic ODE (Christoffel)	analytic ∂M, numeric ODE integ.
Yang et al. (2018)	geodesic energy	autodiff
Park et al. (2023)	none (first-order basis)	autodiff power iteration
Chen et al. (2018)	none (first-order metric)	—
Kwon et al. (2023)	none (first-order)	CLIP-grad + MLP

Table 4: How representative latent-geometry papers obtain second-order quantities. The explicit $/\varepsilon^2$ second difference (Sections 3, 4) is largely avoided in published work, but it is the obvious choice for a practitioner adding a curvature measurement, which is where the hazard bites.

7 Scope and limitations

One limitation matters more than the rest. We show that the measurement can flip, not that any published number is wrong; the audit's point is that the field mostly avoids the trap. The sign inversion of Section 2 is on a ratio ρ that we define to expose method dependence, not a quantity from prior work. The remaining caveats are routine. The non-recoverability result is specific to second-order finite differences; first-order finite differences, including score-JVPs, are well conditioned, as we state. The instrument is algorithmically standard but operationally decisive: composing forward-mode AD is textbook, yet it is what makes exact higher-order pushforward routine where the finite-difference alternative fails. Finally, we show no downstream task win: this is a measurement-correctness result, and whether accurate second-order geometry changes downstream conclusions is left to future work.

8 Conclusion

As geometric analysis of generative models moves from first-order bases toward curvature and geodesics, how one takes the second derivative stops being a detail. Our measurements argue for taking it with composed forward-mode AD rather than finite differences: the finite-difference second derivative of a deep generator has no usable step in single precision and no rescue in double, and its error is large enough to flip a measured sign, whereas the exact instrument costs one extra forward pass. The methods examined here stay first-order or energy-based and so avoid the problem today. Whether that holds as the field reaches for the second-order quantities the instrument now makes cheap to compute exactly is the open question this work is meant to flag.

Reproducibility statement

Upon acceptance we will release the exact-instrument module, the analytic referee, and the StyleGAN and Stable-Diffusion scripts, with fixed seeds and the `metrics.json` behind every table; the public code link is withheld during anonymous review. The generators are the public StyleGAN-bedroom-256 weights (Karras et al., 2019) and Stable Diffusion v1.5 (Rombach et al., 2022) with a 20-step DDIM schedule (Song et al., 2021); all experiments run on a single 8 GB GPU.

References

Georgios Arvanitidis, Lars Kai Hansen, and Søren Hauberg. Latent space oddity: on the curvature of deep generative models. In *International Conference on Learning Representations (ICLR)*, 2018. arXiv:1710.11379.

Nutan Chen, Alexej Klushyn, Richard Kurle, Xueyan Jiang, Justin Bayer, and Patrick van der Smagt. Metrics for deep generative models. In *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2018. arXiv:1711.01204.

Andreas Griewank and Andrea Walther. *Evaluating Derivatives: Principles and Techniques of Algorithmic Differentiation*. SIAM, 2nd edition, 2008.

Nicholas J. Higham. *Accuracy and Stability of Numerical Algorithms*. SIAM, 2nd edition, 2002.

Jonathan Ho and Tim Salimans. Classifier-free diffusion guidance. *arXiv preprint arXiv:2207.12598*, 2022.

Tero Karras, Samuli Laine, and Timo Aila. A style-based generator architecture for generative adversarial networks. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019. arXiv:1812.04948.

Tero Karras, Samuli Laine, Miika Aittala, Janne Hellsten, Jaakko Lehtinen, and Timo Aila. Analyzing and improving the image quality of stylegan. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020. arXiv:1912.04958.

Mingi Kwon, Jaeseok Jeong, and Youngjung Uh. Diffusion models already have a semantic latent space. In *International Conference on Learning Representations (ICLR)*, 2023. arXiv:2210.10960.

Yong-Hyun Park, Mingi Kwon, Jaewoong Choi, Junghyo Jo, and Youngjung Uh. Understanding the latent space of diffusion models through the lens of riemannian geometry. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. arXiv:2307.12868.

Adam Paszke, Sam Gross, Francisco Massa, et al. Pytorch: An imperative style, high-performance deep learning library. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

William H. Press, Saul A. Teukolsky, William T. Vetterling, and Brian P. Flannery. *Numerical Recipes: The Art of Scientific Computing*. Cambridge University Press, 3rd edition, 2007.

Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022. arXiv:2112.10752.

Shinnosuke Saito and Takashi Matsubara. Be tangential to manifold: Discovering riemannian metric for diffusion models. *arXiv preprint arXiv:2510.05509*, 2025a.

Shinnosuke Saito and Takashi Matsubara. Image interpolation with score-based riemannian metrics of diffusion models. *arXiv preprint arXiv:2504.20288*, 2025b.

Hang Shao, Abhishek Kumar, and P. Thomas Fletcher. The riemannian geometry of deep generative models. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, 2018. arXiv:1711.08014.

Jiaming Song, Chenlin Meng, and Stefano Ermon. Denoising diffusion implicit models. In *International Conference on Learning Representations (ICLR)*, 2021. arXiv:2010.02502.

Tao Yang, Georgios Arvanitidis, Dongmei Fu, Xiaogang Li, and Søren Hauberg. Geodesic clustering in deep generative models. *arXiv preprint arXiv:1809.04747*, 2018.

A Stable-Diffusion experimental setup

The sign-inversion experiment of Section 2 uses Stable Diffusion v1.5 (Rombach et al., 2022) at 256×256 with a 20-step DDIM schedule (Song et al., 2021) and classifier-free guidance (Ho & Salimans, 2022). The h-space is the bottleneck activation of the U-Net mid-block, of shape $(1, 1280, 4, 4)$. For a base latent we run the deterministic DDIM chain to the edit step $t_{\text{edit}} = 12$, then study the map $\alpha \mapsto G(\alpha)$, where $G(\alpha)$ is the decoded output image obtained by injecting a perturbation $\alpha\, v$ into h-space at step t_{edit} and completing the chain. The prompt is fixed ("a photograph of a face"), and all randomness is seeded.

We draw $K = 16$ unit-norm h-space probes, build the Jacobian restricted to their span column by column as Je_k, take its SVD, and keep the top $k = 12$ right-singular vectors as edit directions u_i with singular values σ_i. Along each u_i we form the curvature ratio $\rho_i = |\partial_\alpha^2 G|/|\partial_\alpha G|$ at $\alpha = 0$, with $|\cdot|$ the mean absolute value over output pixels. We report Spearman(σ, ρ) over the 12 directions, for each of 6 seeds (0 through 5). The finite-difference estimate of the inner second derivative uses a central difference of the JVP at $\varepsilon \in \{0.1, 0.05, 0.01, 0.001\}$; the main text reports $\varepsilon = 0.05$ and Appendix F gives all four. Every run fits in 4.6 GB of GPU memory.

B The composed-jvp instrument

Forward-mode AD returns $\partial f(x)[v]$ alongside $f(x)$ in one pass. Nesting it gives the exact second-order pushforward with no step size:

```
def pushforward2(G, x, u):       # exact d^2/d alpha^2 G(x + alpha u) at alpha=0
    f  = lambda a: G(x + a * u)              # curve through x in direction u
    df = lambda a: jvp(f, (a,), (1.0,))[1]   # first-order pushforward, exact
    _, d2 = jvp(df, (0.0,), (1.0,))          # directional derivative of df, exact
    return d2                                # second-order pushforward
```

Here `jvp` is `torch.func.jvp`. The inner call computes $\partial_\alpha G$; the outer call differentiates it again along the same tangent. No Jacobian or Hessian is materialized, so peak memory is that of a single forward evaluation of G, independent of the output dimension $3HW$. The k-th order pushforward follows from k nested calls. For the diffusion experiments G is the DDIM chain from t_{edit} onward; the same wrapper runs unmodified on the StyleGAN synthesis network.

C Finite-difference error: derivation

Second derivative. Taylor expansion about α gives $f(\alpha \pm \varepsilon) = f \pm \varepsilon f' + \frac{\varepsilon^2}{2} f'' \pm \frac{\varepsilon^3}{6} f''' + \frac{\varepsilon^4}{24} f^{(4)} \pm \cdots$, so the odd terms cancel in the sum and

$$\frac{f(\alpha+\varepsilon) - 2f(\alpha) + f(\alpha-\varepsilon)}{\varepsilon^2} = f''(\alpha) + \frac{\varepsilon^2}{12} f^{(4)}(\alpha) + O(\varepsilon^4).$$

Each evaluated value carries a rounding error of size $\varepsilon_{\text{mach}}|f|$, and the three-term numerator accumulates $O(\varepsilon_{\text{mach}}|f|)$, divided by ε^2. The total error is therefore $g(\varepsilon) = c_1\varepsilon^2 + c_2\varepsilon_{\text{mach}}/\varepsilon^2$ with $c_1 = |f^{(4)}|/12$ and $c_2 = O(|f|)$. Setting $g'(\varepsilon) = 2c_1\varepsilon - 2c_2\varepsilon_{\text{mach}}/\varepsilon^3 = 0$ gives $\varepsilon^\star = (c_2/c_1)^{1/4}\varepsilon_{\text{mach}}^{1/4}$ and $g(\varepsilon^\star) = 2\sqrt{c_1 c_2}\,\varepsilon_{\text{mach}}^{1/2}$. Dividing by $|f''|$ gives the relative floor $2\sqrt{|f^{(4)}f|}/|f''| \cdot \varepsilon_{\text{mach}}^{1/2}$.

First derivative. The central first difference $\big(f(\alpha+\varepsilon) - f(\alpha-\varepsilon)\big)/2\varepsilon = f' + \frac{\varepsilon^2}{6} f''' + O(\varepsilon^4)$ has truncation $O(\varepsilon^2)$ and roundoff $O(\varepsilon_{\text{mach}}|f|/\varepsilon)$, so $g(\varepsilon) = c_1\varepsilon^2 + c_2\varepsilon_{\text{mach}}/\varepsilon$, minimized at $\varepsilon^\star \sim \varepsilon_{\text{mach}}^{1/3}$ with floor $\varepsilon_{\text{mach}}^{2/3}$. The first-derivative floor ($\varepsilon_{\text{mach}}^{2/3} \approx 2\times10^{-5}$ in fp32) is far below the second-derivative floor ($\varepsilon_{\text{mach}}^{1/2} \approx 3\times10^{-4}$ benign, and orders of magnitude worse once the constant $|f^{(4)}f|/f''^2$ is large), which is the asymmetry the paper rests on.

D StyleGAN sweep: full table

Table 3 reports a subset of steps. Table 5 gives the complete fp32 sweep over 6 directions (3 attributes `indoor_lighting/wood/view` and 3 random) ×16 samples. The exact curvature is 1.17.

step δ	$\lvert D_\delta^2 G\rvert$	2nd-order rel. err.	1st-order rel. err.
8	7.07×10^{-3}	74.5%	63.81%
5	1.38×10^{-2}	57.3%	54.01%
3	2.65×10^{-2}	45.2%	44.07%
2	4.32×10^{-2}	56.1%	37.12%
1.5	6.09×10^{-2}	73.6%	32.39%
1	9.92×10^{-2}	106.4%	26.22%
0.7	1.50×10^{-1}	142.0%	21.34%
0.5	2.17×10^{-1}	184.2%	17.36%
0.3	3.68×10^{-1}	266.9%	12.50%
0.2	5.56×10^{-1}	355.2%	9.37%
0.1	1.06	622.8%	5.47%
0.05	1.86	1429.3%	3.05%
0.02	4.05	6772.5%	2.44%
0.01	8.74	25023.0%	5.71%
0.005	2.36×10^{1}	93346.3%	15.04%
0.001	4.09×10^{2}	1985856.3%	139.11%

Table 5: Full fp32 StyleGAN sweep. The second-order relative error is U-shaped with a floor of 45.2% at $\delta = 3$; the first-order error has a clean minimum of 2.44% at $\delta = 0.02$, confirming the pipeline.

E Precision control: full fp32 vs fp64 sweep

Table 6 gives the per-step second-order relative error in fp32 and fp64 over 3 directions ×8 samples. At large steps the two precisions agree (truncation). At small steps fp32 diverges through catastrophic cancellation while fp64 does not, yet fp64 does not converge either: it plateaus near 3000%, because the chord-scale second difference of a non-smooth (LeakyReLU) generator does not approach the pointwise pushforward. The best-step floor is 47.1% in both.

step δ	fp32 2nd-order rel. err.	fp64 2nd-order rel. err.
3	47.1%	47.1%
1	96.8%	96.8%
0.3	216.8%	216.4%
0.1	431.1%	410.2%
0.03	1467.2%	748.0%
0.01	10557.7%	1203.0%
0.003	1.09×10^{5}%	1815.0%
0.001	8.35×10^{5}%	2373.4%
10^{-4}	6.59×10^{7}%	3183.4%
10^{-5}	5.58×10^{9}%	3470.7%
10^{-6}	4.67×10^{11}%	1502.6%

Table 6: Full precision control. fp64 removes cancellation (small-step columns drop by orders of magnitude) but the floor is unchanged: no precision recovers the second-order pushforward on a real generator.

F Stable Diffusion: per-seed, per-step

Table 7 gives Spearman(σ, ρ) for every seed and every finite-difference step, with the exact column for reference. The exact estimate is negative on five of six seeds. The finite-difference estimate has no step that reproduces it: at each step the across-seed mean is near zero (-0.08 to -0.34) and the sign varies seed to seed.

seed	exact	FD@0.1	FD@0.05	FD@0.01	FD@0.001
0	−0.853	+0.287	+0.182	−0.259	+0.692
1	−0.825	−0.378	−0.552	−0.287	+0.049
2	−0.916	−0.853	−0.783	−0.902	−0.916
3	−0.944	−0.168	−0.399	−0.028	−0.538
4	−0.028	+0.238	+0.552	+0.000	+0.280
5	−0.839	+0.021	−0.070	−0.580	−0.021
mean	−0.734	−0.142	−0.178	−0.343	−0.076

Table 7: Per-seed Spearman(σ, ρ) on Stable Diffusion. No finite-difference step recovers the exact column's consistent negative sign.

G Literature audit: classification criteria

For each paper in Table 4 we asked three questions from the primary source. (i) Does it compute a second-order quantity at all (curvature, a geodesic, the metric derivative / Christoffel symbols)? (ii) If so, is that quantity obtained by an explicit $/\varepsilon^2$ finite difference, by automatic differentiation, by an analytic expression, or as the exact gradient of a discretized energy? (iii) Does a reported conclusion depend on it? A paper is "exposed" only when a second-order quantity is formed by an explicit second-order finite difference. By this test: Shao et al. (2018) discretizes the geodesic energy and its gradient is the discrete Laplacian $g_{i+1} - 2g_i + g_{i-1}$, the exact gradient of $\sum \|g_{i+1} - g_i\|^2$, with no division by ε^2, so it is not exposed; Saito & Matsubara (2025a) uses first-order score differences; Arvanitidis et al. (2018) forms the metric derivative analytically and integrates the geodesic ODE numerically; Yang et al. (2018) differentiates the energy with autodiff; Park et al. (2023), Chen et al. (2018), and Kwon et al. (2023) never take a second derivative. The single exposed published practice is the first-order finite difference of a network derivative, which the asymmetry of Section 3 places on the safe side.

H Reproduction

All experiments run on a single 8 GB GPU with PyTorch 2.2.2 and `numpy<2`. The analytic referee (Table 2) is a CPU script and is deterministic. The StyleGAN sweeps use the public StyleGAN-bedroom-256 weights; the diffusion experiments use Stable Diffusion v1.5. Representative commands:

```
# StyleGAN FD-vs-exact curvature sweep (Table 3 / Table 5, Appendix D)
python higan_dev/scripts/29_shao_fd_vs_exact.py \
    --attrs indoor_lighting wood view --n-random-dirs 3 --num-samples 16

# fp32 vs fp64 precision control (Table 6, Appendix E)  -- 3 dirs x 8 samples
python higan_dev/scripts/30_fp64_precision_control.py --num-samples 8

# Stable-Diffusion sign inversion (Table 1 / Table 7, Appendix F)
python paper/experiments/diffusion/run_park_fd_vs_exact.py --n-seeds 6 \
    --K-probes 16 --top-k 12 --steps 20 --t-edit 12 --resolution 256
```

Each table's `metrics.json` is released alongside the code. The repository link is withheld for anonymous review.